Day 1: Automated modelbuilding and postprocessing - 2D3D + External forcings file conversion

This installation guide prepares you for one of the breakout sessions of the Delft3D User Days - Day 1 about "Automated modelbuilding and postprocessing: 2D3D" and "External forcings file conversion". In this session we will do some hands-on work with Python. The below installation guide provides you with a Python environment called `dfm_tools_env` that contains the packages `dfm_tools` and `HYDROLIB-core` including all their dependencies. With this environment you can run all the example notebooks provided for this breakout session. You can also use the same environment for the breakout session "Deep-dive in pre- and postprocessing" on Delft3D User Days - Day 3, since it also contains the packages `HYDROLIB-core` and `xugrid`. For those who were also present in previous years, the contents of the breakout sessions again include some additions since the packages have many new features and stability improvements.

Installation of Python environment

- Download Miniforge3 from conda-forge.org and install it with the recommended settings
- Open "Miniforge Prompt" (for instance from your windows Start Menu)
- `conda create --name dfm_tools_env python=3.12 -y` (python 3.10 to 3.13 is supported)
- `conda activate dfm_tools_env`
- `python -m pip install dfm_tools[examples]` (version 0.42.0 or higher is required)

Download course materials from github

Click on the download button () for the following notebooks:

- modelbuilder notebook: https://github.com/Deltares/dfm_tools/blob/main/docs/notebooks/modelbuilder_example.ipynb
- post-processing notebook: https://github.com/Deltares/dfm_tools/blob/main/docs/notebooks/postprocessing_example.ipynb

Register to retrieve CMEMS and ECMWF/CDS data (for dfm_tools modelbuilder)

- We will make use of publicly available data from Copernicus Programme of the European Union. To access this data you need to make two accounts at Copernicus Marine Service for CMEMS data and the ECMWF Climate Data Store for ERA5 data.
- Do not forget to accept the license agreements, for ECMWF also accept the dataset license on this page
- You will be prompted for the CMEMS credentials and CDS/ECMWF apikey by the dfm_tools 2D3D modelbuilder notebook.

Doing the python exercises

- Open "Miniforge Prompt" in the folder where you downloaded the example notebooks
- `conda activate dfm_tools_env`
- Launch jupyter with: `python -m jupyter notebook`
- Open the notebook you want to work in and run it with SHIFT+ENTER or press the play button in the toolbar above
- Alternatively you can run the notebooks in VSCode (with the `dfm_tools_env` environment activated)

Installation of DeltaShell GUI (Delft3D FM Suite) (optional)

- If you want to run your newly generated model immediately, you need an FM executable. This is available after installing "Delft3D FM Suite 2026.01" (or a different version)
- Provided via download portal: https://download.deltares.nl/delft3d-user-days-2025-day-1-break-out-session-modelbuilding-and-postprocessing (contains msi and license manager, login using MyDeltares account)
- You will receive a temporary license file via email